MPU-6500
Register Map and Descriptions
Revision 2.1

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1 Revision History

Revision Date	Revision	Description
06/29/2012	1.0	Initial Release
07/15/2013	2.0	Updated Release
09/16/2013	2.1	Updated sections 3 and 4

2 Purpose and Scope

This document provides preliminary information regarding the register map and descriptions for the Motion Processing Unit™ MPU-6500™. This document should be used in conjunction with the MPU-6500 Product Specification (PS-MPU-6500A-00) for detailed features, specifications, and other product information.

3 Register Map

The MPU-6500 has two modes of operation in terms of device programmability:

- MPU-6500 mode
  - This is the default manner of operation.
- MPU-6050 compatible mode
  - If this mode is desired, please contact your local InvenSense representative as this would require a change in the device trim setting.

The following table lists the register map for the MPU-6500 mode. All the registers affected by the differences between the two modes are marked by an asterisk (*) in the register name column in the table below and are explained separately below.

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
00	0	SELF_TEST_X_GYRO	R/W	XG_ST_DATA[7:0]							
01	1	SELF_TEST_Y_GYRO	R/W	YG_ST_DATA[7:0]							
02	2	SELF_TEST_Z_GYRO	R/W	ZG_ST_DATA[7:0]							
0D	13	SELF_TEST_X_ACCEL	R/W	XA_ST_DATA [7:0]							
0E	14	SELF_TEST_Y_ACCEL	R/W	YA_ST_DATA [7:0]							
0F	15	SELF_TEST_Z_ACCEL	R/W	ZA_ST_DATA [7:0]							
13	19	XG_OFFSET_H	R/W	X_OFFS_USR [15:8]							
14	20	XG_OFFSET_L	R/W	X_OFFS_USR [7:0]							
15	21	YG_OFFSET_H	R/W	Y_OFFS_USR [15:8]							
16	22	YG_OFFSET_L	R/W	Y_OFFS_USR [7:0]							
17	23	ZG_OFFSET_H	R/W	Z_OFFS_USR [15:8]							
18	24	ZG_OFFSET_L	R/W	Z_OFFS_USR [7:0]							
19	25	SMPLRT_DIV	R/W	SMPLRT_DIV[7:0]							
1A	26	CONFIG	R/W	-	FIFO_MODE	EXT_SYNC_SET[2:0]			DLPF_CFG[2:0]		
1B	27	GYRO_CONFIG	R/W	XG_ST	YG_ST	ZG_ST	GYRO_FS_SEL [1:0]		-	FCHOICE_B[1:0]	
1C	28	ACCEL_CONFIG	R/W	XA_ST	YA_ST	ZA_ST	ACCEL_FS_SEL[1:0]		-		
1D	29	ACCEL_CONFIG 2	R/W	-				ACCEL_FCHOICE_B	A_DLPF_CFG		
1E	30	LP_ACCEL_ODR *	R/W	-				LPOSC_CLKSEL [3:0]			
1F	31	WOM_THR	R/W	WOM_THRESHOLD [7:0]							
23	35	FIFO_EN	R/W	TEMP_FIFO_EN	GYRO_XOUT	GYRO_YOUT	GYRO_ZOUT	ACCEL	SLV2	SLV1	SLV0
24	36	I2C_MST_CTRL	R/W	MULT_MST_EN	WAIT_FOR_ES	SLV_3_FIFO_EN	I2C_MST_P_NSR	I2C_MST_CLK[3:0]			
25	37	I2C_SLV0_ADDR	R/W	I2C_SLV0_RNW	I2C_ID_0 [6:0]						
26	38	I2C_SLV0_REG	R/W	I2C_SLV0_REG[7:0]							
27	39	I2C_SLV0_CTRL	R/W	I2C_SLV0_EN	I2C_SLV0_BYTE_SW	I2C_SLV0_REG_DIS	I2C_SLV0_GRP	I2C_SLV0_LENG[3:0]			
28	40	I2C_SLV1_ADDR	R/W	I2C_SLV1_RNW	I2C_ID_1 [6:0]						
29	41	I2C_SLV1_REG	R/W	I2C_SLV1_REG[7:0]							
2A	42	I2C_SLV1_CTRL	R/W	I2C_SLV1_EN	I2C_SLV1_BYTE_SW	I2C_SLV1_REG_DIS	I2C_SLV1_GRP	I2C_SLV1_LENG[3:0]			
2B	43	I2C_SLV2_ADDR	R/W	I2C_SLV2_RNW	I2C_ID_2 [6:0]						
2C	44	I2C_SLV2_REG	R/W	I2C_SLV2_REG[7:0]							
2D	45	I2C_SLV2_CTRL	R/W	I2C_SLV2_EN	I2C_SLV2_BYTE_SW	I2C_SLV2_REG_DIS	I2C_SLV2_GRP	I2C_SLV2_LENG[3:0]			

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
2E	46	I2C_SLV3_ADDR	R/W	I2C_SLV3_RNW	I2C_ID_3 [6:0]						
2F	47	I2C_SLV3_REG	R/W	I2C_SLV3_REG[7:0]							
30	48	I2C_SLV3_CTRL	R/W	I2C_SLV3_EN	I2C_SLV3_BYTE_SW	I2C_SLV3_REG_DIS	I2C_SLV3_GRP	I2C_SLV3_LENG [3:0]			
31	49	I2C_SLV4_ADDR	R/W	I2C_SLV4_RNW	I2C_ID_4 [6:0]						
32	50	I2C_SLV4_REG	R/W	I2C_SLV4_REG[7:0]							
33	51	I2C_SLV4_DO	R/W	I2C_SLV4_DO[7:0]							
34	52	I2C_SLV4_CTRL	R/W	I2C_SLV4_EN	SLV4_DONE_INT_EN	I2C_SLV4_REG_DIS	I2C_MST_DLY[4:0]				
35	53	I2C_SLV4_DI	R	I2C_SLV4_DI[7:0]							
36	54	I2C_MST_STATUS	R	PASS_THROUGH	I2C_SLV4_DONE	I2C_LOST_ARB	I2C_SLV4_NACK	I2C_SLV3_NACK	I2C_SLV2_NACK	I2C_SLV1_NACK	I2C_SLV0_NACK
37	55	INT_PIN_CFG	R/W	ACTL	OPEN	LATCH_INT_EN	INT_ANYRD_2CLEAR	ACTL_FSYNC	FSYNC_INT_MODE_EN	BYPASS_EN	-
38	56	INT_ENABLE	R/W	-	WOM_EN	-	FIFO_OFLOW_EN	FSYNC_INT_EN	-	-	RAW_RDY_EN
3A	58	INT_STATUS	R	-	WOM_INT	-	FIFO_OFLOW_INT	FSYNC_INT	-	DMP_INT	RAW_DATA_RDY_INT
3B	59	ACCEL_XOUT_H	R	ACCEL_XOUT_H[15:8]							
3C	60	ACCEL_XOUT_L	R	ACCEL_XOUT_L[7:0]							
3D	61	ACCEL_YOUT_H	R	ACCEL_YOUT_H[15:8]							
3E	62	ACCEL_YOUT_L	R	ACCEL_YOUT_L[7:0]							
3F	63	ACCEL_ZOUT_H	R	ACCEL_ZOUT_H[15:8]							
40	64	ACCEL_ZOUT_L	R	ACCEL_ZOUT_L[7:0]							
41	65	TEMP_OUT_H	R	TEMP_OUT_H[15:8]							
42	66	TEMP_OUT_L	R	TEMP_OUT_L[7:0]							
43	67	GYRO_XOUT_H	R	GYRO_XOUT_H[15:8]							
44	68	GYRO_XOUT_L	R	GYRO_XOUT_L[7:0]							
45	69	GYRO_YOUT_H	R	GYRO_YOUT_H[15:8]							
46	70	GYRO_YOUT_L	R	GYRO_YOUT_L[7:0]							
47	71	GYRO_ZOUT_H	R	GYRO_ZOUT_H[15:8]							
48	72	GYRO_ZOUT_L	R	GYRO_ZOUT_L[7:0]							
49	73	EXT_SENS_DATA_00	R	EXT_SENS_DATA_00[7:0]							
4A	74	EXT_SENS_DATA_01	R	EXT_SENS_DATA_01[7:0]							
4B	75	EXT_SENS_DATA_02	R	EXT_SENS_DATA_02[7:0]							
4C	76	EXT_SENS_DATA_03	R	EXT_SENS_DATA_03[7:0]							
4D	77	EXT_SENS_DATA_04	R	EXT_SENS_DATA_04[7:0]							
4E	78	EXT_SENS_DATA_05	R	EXT_SENS_DATA_05[7:0]							
4F	79	EXT_SENS_DATA_06	R	EXT_SENS_DATA_06[7:0]							
50	80	EXT_SENS_DATA_07	R	EXT_SENS_DATA_07[7:0]							
51	81	EXT_SENS_DATA_08	R	EXT_SENS_DATA_08[7:0]							
52	82	EXT_SENS_DATA_09	R	EXT_SENS_DATA_09[7:0]							
53	83	EXT_SENS_DATA_10	R	EXT_SENS_DATA_10[7:0]							
54	84	EXT_SENS_DATA_11	R	EXT_SENS_DATA_11[7:0]							
55	85	EXT_SENS_DATA_12	R	EXT_SENS_DATA_12[7:0]							
56	86	EXT_SENS_DATA_13	R	EXT_SENS_DATA_13[7:0]							
57	87	EXT_SENS_DATA_14	R	EXT_SENS_DATA_14[7:0]							
58	88	EXT_SENS_DATA_15	R	EXT_SENS_DATA_15[7:0]							
59	89	EXT_SENS_DATA_16	R	EXT_SENS_DATA_16[7:0]							

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
5A	90	EXT_SENS_DATA_17	R	EXT_SENS_DATA_17[7:0]							
5B	91	EXT_SENS_DATA_18	R	EXT_SENS_DATA_18[7:0]							
5C	92	EXT_SENS_DATA_19	R	EXT_SENS_DATA_19[7:0]							
5D	93	EXT_SENS_DATA_20	R	EXT_SENS_DATA_20[7:0]							
5E	94	EXT_SENS_DATA_21	R	EXT_SENS_DATA_21[7:0]							
5F	95	EXT_SENS_DATA_22	R	EXT_SENS_DATA_22[7:0]							
60	96	EXT_SENS_DATA_23	R	EXT_SENS_DATA_23[7:0]							
63	99	I2C_SLV0_DO	R/W	I2C_SLV0_DO[7:0]							
64	100	I2C_SLV1_DO	R/W	I2C_SLV1_DO[7:0]							
65	101	I2C_SLV2_DO	R/W	I2C_SLV2_DO[7:0]							
66	102	I2C_SLV3_DO	R/W	I2C_SLV3_DO[7:0]							
67	103	I2C_MST_DELAY_CTRL	R/W	DELAY_ES_SHADOW	-	-	I2C_SLV4_DLY_EN	I2C_SLV3_DLY_EN	I2C_SLV2_DLY_EN	I2C_SLV1_DLY_EN	I2C_SLV0_DLY_EN
68	104	SIGNAL_PATH_RESET	R/W	-	-	-	-	-	GYRO_RST	ACCEL_RST	TEMP_RST
69	105	ACCEL_INTEL_CTRL	R/W	ACCEL_INTEL_EN	ACCEL_INTEL_MODE	-		-		-	
6A	106	USER_CTRL	R/W	DMP_EN	FIFO_EN	I2C_MST_EN	I2C_IF_DIS	DMP_RST	FIFO_RST	I2C_MST_RST	SIG_COND_RST
6B	107	PWR_MGMT_1	R/W	DEVICE_RESET	SLEEP	CYCLE	GYRO_STANDBY	TEMP_DIS	CLKSEL[2:0]		
6C	108	PWR_MGMT_2 *	R/W	-		DIS_XA	DIS_YA	DIS_ZA	DIS_XG	DIS_YG	DIS_ZG
72	114	FIFO_COUNT_H	R/W	-			FIFO_CNT[12:8]				
73	115	FIFO_COUNT_L	R/W	FIFO_CNT[7:0]							
74	116	FIFO_R_W	R/W	D[7:0]							
75	117	WHO_AM_I	R	WHOAMI[7:0]							
77	119	XA_OFFSET_H *	R/W	XA_OFFS [14:7]							
78	120	XA_OFFSET_L *	R/W	XA_OFFS [6:0]							-
7A	122	YA_OFFSET_H *	R/W	YA_OFFS [14:7]							
7B	123	YA_OFFSET_L *	R/W	YA_OFFS [6:0]							-
7D	125	ZA_OFFSET_H *	R/W	ZA_OFFS [14:7]							
7E	126	ZA_OFFSET_L *	R/W	ZA_OFFS [6:0]							-

Table 1 MPU-6500 mode register map

The table below lists the registers in the MPU-6050 compatible mode register map that are different from the MPU-6500 mode.

Addr (Hex)	Addr (Dec.)	Register Name	Serial I/F	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
6	6	XA_OFFSET_H	R/W	XA_OFFSET [14:7]							
7	7	XA_OFFSET_L	R/W	XA_OFFSET [6:0]							-
8	8	YA_OFFSET_H	R/W	YA_OFFSET [14:7]							
9	9	YA_OFFSET_L	R/W	YA_OFFSET [6:0]							-
A	10	ZA_OFFSET_H	R/W	ZA_OFFSET [14:7]							
B	11	ZA_OFFSET_L	R/W	ZA_OFFSET [6:0]							-
6C	108	PWR_MGMT_2	R/W	LP_WAKE_CTRL[1:0]		DIS_XA	DIS_YA	DIS_ZA	DIS_XG	DIS_YG	DIS_ZG

Table 2 MPU-6050 compatible mode register map

Note: Register Names ending in _H and _L contain the high and low bytes, respectively, of an internal register value.

In the detailed register tables that follow, register names are in capital letters, while register values are in capital letters and italicized. For example, the ACCEL_XOUT_H register (Register 59) contains the 8 most significant bits, *ACCEL_XOUT*[15:8], of the 16-bit X-Axis accelerometer measurement, *ACCEL_XOUT*.

The reset value is 0x00 for all registers other than the registers below, also the self-test registers contain pre-programmed values and will not be 0x00 after reset.

- Register 107 (0x01) Power Management 1
- Register 117 (0x70) WHO_AM_I

4 Register Descriptions

This section describes the function and contents of each register within the MPU-6500. All the descriptions relate to the default MPU-6500 mode of operation. Any differences specific to the MPU-6050 mode of operation are described in the specific register sections below.

Note: The device will come up in active mode upon power-up.

4.1 Registers 0 to 2 – Gyroscope Self-Test Registers

Serial IF: R/W

REGISTER	BITS	FUNCTION
SELF_TEST_X_GYRO	XG_ST_DATA[7:0]	The value in this register indicates the self-test output generated during manufacturing tests. This value is to be used to check against subsequent self-test outputs performed by the end user.
SELF_TEST_Y_GYRO	YG_ST_DATA[7:0]	The value in this register indicates the self-test output generated during manufacturing tests. This value is to be used to check against subsequent self-test outputs performed by the end user.
SELF_TEST_Z_GYRO	ZG_ST_DATA[7:0]	The value in this register indicates the self-test output generated during manufacturing tests. This value is to be used to check against subsequent self-test outputs performed by the end user.

Please refer to the following document for information on Self-Test: *MPU-6500 Accelerometer and Gyroscope Self-Test Implementation; AN-MPU-6500A-02.*

4.2 Registers 13 to 15 – Accelerometer Self-Test Registers

Serial IF: R/W

REGISTER	BITS	FUNCTION
SELF_TEST_X_ACCEL	XA_ST_DATA[7:0]	The value in this register indicates the self-test output generated during manufacturing tests. This value is to be used to check against subsequent self-test outputs performed by the end user.
SELF_TEST_Y_ACCEL	YA_ST_DATA[7:0]	The value in this register indicates the self-test output generated during manufacturing tests. This value is to be used to check against subsequent self-test outputs performed by the end user.
SELF_TEST_Z_ACCEL	ZA_ST_DATA[7:0]	The value in this register indicates the self-test output generated during manufacturing tests. This value is to be used to check against subsequent self-test outputs performed by the end user.

Please refer to the following document for information on Self-Test: *MPU-6500 Accelerometer and Gyroscope Self-Test Implementation; AN-MPU-6500A-02.*

4.3 Registers 19 to 24 – Gyro Offset Registers

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	X_OFFS_USR[15:8]	Bits 15 to 8 of the 16-bit offset of X gyroscope (2's complement). This register is used to remove DC bias from the sensor output. The value in this register is added to the gyroscope sensor value before going into the sensor register.
[7:0]	X_OFFS_USR[7:0]	Bits 7 to 0 of the 16-bit offset of X gyroscope (2's complement). This register is used to remove DC bias from the sensor output. The value in this register is added to the gyroscope sensor value before going into the sensor register.
[7:0]	Y_OFFS_USR[15:8]	Bits 15 to 8 of the 16-bit offset of Y gyroscope (2's complement). This register is used to remove DC bias from the sensor output. The value in this register is added to the gyroscope sensor value before going into the sensor register.
[7:0]	Y_OFFS_USR[7:0]	Bits 7 to 0 of the 16-bit offset of Y gyroscope (2's complement). This register is used to remove DC bias from the sensor output. The value in this register is added to the gyroscope sensor value before going into the sensor register.
[7:0]	Z_OFFS_USR[15:8]	Bits 15 to 8 of the 16-bit offset of Z gyroscope (2's complement). This register is used to remove DC bias from the sensor output. The value in this register is added to the gyroscope sensor value before going into the sensor register.
[7:0]	Z_OFFS_USR[7:0]	Bits 7 to 0 of the 16-bit offset of Z gyroscope (2's complement). This register is used to remove DC bias from the sensor output. The value in this register is added to the gyroscope sensor value before going into the sensor register.

4.4 Register 25 – Sample Rate Divider

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	SMPLRT_DIV[7:0]	Divides the internal sample rate (see register CONFIG) to generate the sample rate that controls sensor data output rate, FIFO sample rate. NOTE: This register is only effective when FCHOICE = 2'b11 (FCHOICE_B register bits are 2'b00), and (0 < DLPF_CFG < 7) This is the update rate of sensor register. SAMPLE_RATE = INTERNAL_SAMPLE_RATE / (1 + SMPLRT_DIV) where INTERNAL_SAMPLE_RATE = 1kHz

Data should be sampled at or above sample rate; SMPLRT_DIV is only used for1kHz internal sampling.

4.5 Register 26 – Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	-	Reserved
[6]	FIFO_MODE	When set to '1', when the FIFO is full, additional writes will not be written to FIFO. When set to '0', when the FIFO is full, additional writes will be written to the FIFO, replacing the oldest data.
[5:3]	EXT_SYNC_SET[2:0]	Enables the FSYNC pin data to be sampled. **EXT_SYNC_SET** / **FSYNC bit location** 0 – function disabled 1 – TEMP_OUT_L[0] 2 – GYRO_XOUT_L[0] 3 – GYRO_YOUT_L[0] 4 – GYRO_ZOUT_L[0] 5 – ACCEL_XOUT_L[0] 6 – ACCEL_YOUT_L[0] 7 – ACCEL_ZOUT_L[0] FSYNC will be latched to capture short strobes. This will be done such that if FSYNC toggles, the latched value toggles, but won't toggle again until the new latched value is captured by the sample rate strobe.
[2:0]	DLPF_CFG[2:0]	For the DLPF to be used, FCHOICE[1:0] must be set to 2'b11, FCHOICE_B[1:0] is 2'b00. See the table below.

The DLPF is configured by *DLPF_CFG,* when *FCHOICE_B* [1:0] = 2b'00. The gyroscope and temperature sensor are filtered according to the value of *DLPF_CFG* and *FCHOICE_B* as shown in the table below. Note that FCHOICE mentioned in the table below is the inverted value of *FCHOICE_B* (e.g. FCHOICE=2b'00 is same as FCHOICE_B=2b'11).

FCHOICE_B		DLPF_CFG	Gyroscope			Temperature Sensor	
<1>	<0>		Bandwidth (Hz)	Delay (ms)	Fs (kHz)	Bandwidth (Hz)	Delay (ms)
x	1	x	8800	0.064	32	4000	0.04
1	0	x	3600	0.11	32	4000	0.04
0	0	0	250	0.97	8	4000	0.04
0	0	1	184	2.9	1	188	1.9
0	0	2	92	3.9	1	98	2.8
0	0	3	41	5.9	1	42	4.8
0	0	4	20	9.9	1	20	8.3
0	0	5	10	17.85	1	10	13.4
0	0	6	5	33.48	1	5	18.6
0	0	7	3600	0.17	8	4000	0.04

4.6 Register 27 – Gyroscope Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	XG_ST	X Gyro self-test
[6]	YG_ST	Y Gyro self-test
[5]	ZG_ST	Z Gyro self-test
[4:3]	GYRO_FS_SEL[1:0]	Gyro Full Scale Select: 00 = ±250dps 01= ±500dps 10 = ±1000dps 11 = ±2000dps
[2]	-	Reserved
[1:0]	FCHOICE_B[1:0]	Used to bypass DLPF as shown in the table above.

4.7 Register 28 – Accelerometer Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	XA_ST	X Accel self-test
[6]	YA_ST	Y Accel self-test

[5]	ZA_ST	Z Accel self-test
[4:3]	ACCEL_FS_SEL[1:0]	Accel Full Scale Select: ±2g (00), ±4g (01), ±8g (10), ±16g (11)
[2:0]	-	Reserved

4.8 Register 29 – Accelerometer Configuration 2

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:4]	-	Reserved
[3]	ACCEL_FCHOICE_B	Used to bypass DLPF as shown in the table below.
[2:0]	A_DLPF_CFG	Accelerometer low pass filter setting as shown in the second table below.

Accelerometer Data Rates and Bandwidths (Normal Mode)

ACCEL_FCHOICE_B	A_DLPF_CFG	Output			
		Bandwidth (Hz)	Delay (ms)	Noise Density (ug/rtHz)	Rate (kHz)
1	X	1.13 K	0.75	220	4
0	0	460	1.94	220	1
0	1	184	5.80	220	1
0	2	92	7.80	220	1
0	3	41	11.80	220	1
0	4	20	19.80	220	1
0	5	10	35.70	220	1
0	6	5	66.96	220	1
0	7	460	1.94	220	1

The data output rate of the DLPF filter block can be further reduced by a factor of 1/(1+SMPLRT_DIV), where SMPLRT_DIV is an 8-bit integer. Following is a small subset of ODRs that are configurable for the accelerometer in the normal mode in this manner (Hz):

3.91, 7.81, 15.63, 31.25, 62.50, 125, 250, 500, 1K

The following table lists the approximate accelerometer filter bandwidths available in the low-power mode of operation.

In the low-power mode of operation, the accelerometer is duty-cycled. ACCEL_FCHOICE=0 for all options.

Accelerometer Data Rates and Bandwidths (Low-Power Mode)

ACCEL_FCHOICE	ODR (Hz)	Output	
		Bandwidth (Hz)	Delay (ms)
0	0.24	1.1 k	1
0	0.49	1.1 k	1
0	0.98	1.1 k	1
0	1.95	1.1 k	1
0	3.91	1.1 k	1
0	7.81	1.1 k	1
0	15.63	1.1 k	1
0	31.25	1.1 k	1
0	62.50	1.1 k	1
0	125	1.1 k	1
0	250	1.1 k	1
0	500	1.1 k	1

As you can see from the tables above, some of the ODRs can be configured in the normal accelerometer mode as well as low power mode.

For further details on how to configure the individual ODRs, please refer to register 30 Low Power Accelerometer ODR Control.

4.9 Register 30 – Low Power Accelerometer ODR Control

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:4]	-	Reserved
[3:0]	LPOSC_CLKSEL[3:0]	Sets the frequency of waking up the chip to take a sample of accel data – the low power accel Output Data Rate.

LPOSC_CLKSEL	Output Frequency (Hz)
0	0.24
1	0.49
2	0.98
3	1.95
4	3.91
5	7.81
6	15.63
7	31.25
8	62.50
9	125
10	250
11	500
12-15	Reserved

Please refer to the register 108 Power Management 2 for further details on the MPU-6050 compatible mode.

4.10 Register 31 – Wake-on Motion Threshold

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	WOM_THRESHOLD	This register holds the threshold value for the Wake on Motion Interrupt for accel x/y/z axes. LSB = 4mg. Range is 0mg to 1020mg.

For more details on how to configure the Wake-on-Motion interrupt, please refer to section 5 in the MPU-6500 Product Specification document.

4.11 Register 35 – FIFO Enable

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	TEMP_OUT	1 – Write TEMP_OUT_H and TEMP_OUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled
[6]	GYRO_XOUT	1 – Write GYRO_XOUT_H and GYRO_XOUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled
[5]	GYRO_YOUT	1 – Write GYRO_YOUT_H and GYRO_YOUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled NOTE: Enabling any one of the bits corresponding to the Gyros or Temp data paths, data is buffered into the FIFO even though that data path is not enabled.
[4]	GYRO_ZOUT	1 – Write GYRO_ZOUT_H and GYRO_ZOUT_L to the FIFO at the sample rate; If enabled, buffering of data occurs even if data path is in standby. 0 – function is disabled
[3]	ACCEL	1 – write ACCEL_XOUT_H, ACCEL_XOUT_L, ACCEL_YOUT_H, ACCEL_YOUT_L, ACCEL_ZOUT_H, and ACCEL_ZOUT_L to the FIFO at the sample rate; 0 – function is disabled
[2]	SLV_2	1 – write EXT_SENS_DATA registers associated to SLV_2 (as determined by I2C_SLV0_CTRL, I2C_SLV1_CTRL, and I2C_SL20_CTRL) to the FIFO at the sample rate; 0 – function is disabled
[1]	SLV_1	1 – write EXT_SENS_DATA registers associated to SLV_1 (as determined by I2C_SLV0_CTRL and I2C_SLV1_CTRL) to the FIFO at the sample rate; 0 – function is disabled
[0]	SLV_0	1 – write EXT_SENS_DATA registers associated to SLV_0 (as determined by I2C_SLV0_CTRL) to the FIFO at the sample rate; 0 – function is disabled NOTE: See I2C_SLV3_CTRL register to enable this feature for SLV_3

Note: For further information regarding the association of EXT_SENS_DATA registers to particular slave devices, please refer to Registers 73 to 96.

4.12 Register 36 – I2C Master Control

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	MULT_MST_EN	Enables multi-master capability. When disabled, clocking to the I2C_MST_IF can be disabled when not in use and the logic to detect lost arbitration is disabled.
[6]	WAIT_FOR_ES	Delays the data ready interrupt until external sensor data is loaded. If I2C_MST_IF is disabled, the interrupt will still occur.
[5]	SLV_3_FIFO_EN	1 – write EXT_SENS_DATA registers associated to SLV_3 (as determined by I2C_SLV0_CTRL and I2C_SLV1_CTRL and I2C_SLV2_CTRL) to the FIFO at the sample rate; 0 – function is disabled
[4]	I2C_MST_P_NSR	This bit controls the I2C Master's transition from one slave read to the next slave read. If 0, there is a restart between reads. If 1, there is a stop between reads.
[3:0]	I2C_MST_CLK [3:0]	*I2C_MST_CLK* is a 4 bit unsigned value which configures a divider on the MPU-6500 internal 8MHz clock. It sets the I^2C master clock speed according to the following table:

I2C_MST_CLK	I^2C Master Clock Speed	8MHz Clock Divider
0	348 kHz	23
1	333 kHz	24
2	320 kHz	25
3	308 kHz	26
4	296 kHz	27
5	286 kHz	28
6	276 kHz	29
7	267 kHz	30
8	258 kHz	31
9	500 kHz	16
10	471 kHz	17
11	444 kHz	18
12	421 kHz	19
13	400 kHz	20
14	381 kHz	21
15	364 kHz	22

Note: For further information regarding the association of EXT_SENS_DATA registers to particular slave devices, please refer to Registers 73 to 96.

4.13 Registers 37 to 39 – I^2C Slave 0 Control

Register 37 - I2C_SLV0_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV0_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_0[6:0]	Physical address of I2C slave 0

Register 38 - I2C_SLV0_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV0_REG[7:0]	I2C slave 0 register address from where to begin data transfer

Register 39 - I2C_SLV0_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV0_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register, which is always EXT_SENS_DATA_00 for I2C slave 0. 0 – function is disabled for this slave

BIT	NAME	FUNCTION
[6]	I2C_SLV0_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV0_REG[0] = 1, or if the last byte read has a register address lsb = 0. For example, if I2C_SLV0_REG = 0x1, and I2C_SLV0_LENG = 0x4: 1) The first byte read from address 0x1 will be stored at EXT_SENS_DATA_00, 2) the second and third bytes will be read and swapped, so the data read from address 0x2 will be stored at EXT_SENS_DATA_02, and the data read from address 0x3 will be stored at EXT_SENS_DATA_01, 3) The last byte read from address 0x4 will be stored at EXT_SENS_DATA_03 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV0_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4]	I2C_SLV0_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc.., or if the groups are address 1 and 2, 3 and 4, etc.. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV0_LENG[3:0]	Number of bytes to be read from I2C slave 0

4.14 Registers 40 to 42 – I^2C Slave 1 Control

Register 40 - I2C_SLV1_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV1_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_1[6:0]	Physical address of I2C slave 1

Register 41 - I2C_SLV1_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV1_REG[7:0]	I2C slave 1 register address from where to begin data transfer

Register 42 - I2C_SLV1_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV1_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register as determined by I2C_SLV1_EN and I2C_SLV1_LENG. 0 – function is disabled for this slave

BIT	NAME	FUNCTION
[6]	I2C_SLV1_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV1_REG[0] = 1, or if the last byte read has a register address lsb = 0. For example, if I2C_SLV1_EN = 0x1, and I2C_SLV1_LENG = 0x3 (to show swap has to do with I2C slave address not EXT_SENS_DATA address), and if I2C_SLV1_REG = 0x1, and I2C_SLV1_LENG = 0x4: 1) The first byte read from address 0x1 will be stored at EXT_SENS_DATA_03 (slave 0's data will be in EXT_SENS_DATA_00, EXT_SENS_DATA_01, and EXT_SENS_DATA_02), 2) the second and third bytes will be read and swapped, so the data read from address 0x2 will be stored at EXT_SENS_DATA_04, and the data read from address 0x3 will be stored at EXT_SENS_DATA_05, 3) The last byte read from address 0x4 will be stored at EXT_SENS_DATA_06 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV1_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4]	I2C_SLV1_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc.., or if the groups are address 1 and 2, 3 and 4, etc. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV1_LENG[3:0]	Number of bytes to be read from I2C slave 1

4.15 Registers 43 to 45 – I^2C Slave 2 Control

Register 43 - I2C_SLV2_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV2_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_2[6:0]	Physical address of I2C slave 2

Register 44 - I2C_SLV2_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV2_REG[7:0]	I2C slave 2 register address from where to begin data transfer.

Register 45 - I2C_SLV2_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV2_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register as determined by I2C_SLV0_EN, I2C_SLV0_LENG, I2C_SLV1_EN and I2C_SLV1_LENG. 0 – function is disabled for this slave
[6]	I2C_SLV2_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV2_REG[0] = 1, or if the last byte read has a register address LSB = 0. See I2C_SLV1_CTRL for an example. 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV2_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data

BIT	NAME	FUNCTION
[4]	I2C_SLV2_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc., or if the groups are address 1 and 2, 3 and 4, etc. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV2_LENG[3:0]	Number of bytes to be read from I2C slave 2

4.16 Registers 46 to 48 – I²C Slave 3 Control

Register 46 - I2C_SLV3_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV3_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_3[6:0]	Physical address of I2C slave 3

Register 47 - I2C_SLV3_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV3_REG[7:0]	I2C slave 3 register address from where to begin data transfer

Register 48 - I2C_SLV3_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV3_EN	1 – Enable reading data from this slave at the sample rate and storing data at the first available EXT_SENS_DATA register as determined by I2C_SLV0_EN, I2C_SLV0_LENG, I2C_SLV1_EN, I2C_SLV1_LENG, I2C_SLV2_EN and I2C_SLV2_LENG. 0 – function is disabled for this slave
[6]	I2C_SLV3_BYTE_SW	1 – Swap bytes when reading both the low and high byte of a word. Note there is nothing to swap after reading the first byte if I2C_SLV3_REG[0] = 1, or if the last byte read has a register address LSB = 0. See I2C_SLV1_CTRL for an example. 0 – no swapping occurs, bytes are written in order read.
[5]	I2C_SLV0_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4]	I2C_SLV3_GRP	External sensor data typically comes in as groups of two bytes. This bit is used to determine if the groups are from the slave's register address 0 and 1, 2 and 3, etc.., or if the groups are address 1 and 2, 3 and 4, etc. 0 indicates slave register addresses 0 and 1 are grouped together (odd numbered register ends the group). 1 indicates slave register addresses 1 and 2 are grouped together (even numbered register ends the group). This allows byte swapping of registers that are grouped starting at any address.
[3:0]	I2C_SLV3_LENG[3:0]	Number of bytes to be read from I2C slave 3

4.17 Registers 49 to 53 – I^2C Slave 4 Control

Register 49 - I2C_SLV4_ADDR

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV4_RNW	1 – Transfer is a read 0 – Transfer is a write
[6:0]	I2C_ID_4[6:0]	Physical address of I2C slave 4

Register 50 - I2C_SLV4_REG

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV4_REG[7:0]	I2C slave 4 register address from where to begin data transfer

Register 51 - I2C_SLV4_DO

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV4_DO[7:0]	Data to be written to I2C Slave 4 if enabled.

Register 52 - I2C_SLV4_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	I2C_SLV4_EN	1 – Enable data transfer with this slave at the sample rate. If read command, store data in I2C_SLV4_DI register, if write command, write data stored in I2C_SLV4_DO register. Bit is cleared when a single transfer is complete. Be sure to write I2C_SLV4_DO first 0 – function is disabled for this slave
[6]	SLV4_DONE_INT_EN	1 – Enables the completion of the I2C slave 4 data transfer to cause an interrupt. 0 – Completion of the I2C slave 4 data transfer will not cause an interrupt.
[5]	I2C_SLV4_REG_DIS	When set, the transaction does not write a register value, it will only read data, or write data
[4:0]	I2C_MST_DLY	When enabled via the I2C_MST_DELAY_CTRL, those slaves will only be enabled every (1+I2C_MST_DLY) samples (as determined by the SMPLRT_DIV and DLPF_CFG registers.

Register 53 - I2C_SLV4_DI

Serial IF: R

Reset value: 0x00

BIT	NAME	FUNCTION

BIT	NAME	FUNCTION
[7:0]	I2C_SLV4_DI[7:0]	Data read from I2C Slave 4.

4.18 Register 54 – I^2C Master Status

Serial IF: R/C

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	PASS_THROUGH	Status of FSYNC interrupt – used as a way to pass an external interrupt through this chip to the host. If enabled in the INT_PIN_CFG register by asserting bit FSYNC_INT_EN and if the FSYNC signal transitions from low to high, this will cause an interrupt. A read of this register clears all status bits in this register.
[6]	I2C_SLV4_DONE	Asserted when I2C slave 4's transfer is complete, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted, and if the SLV4_DONE_INT_EN bit is asserted in the I2C_SLV4_CTRL register.
[5]	I2C_LOST_ARB	Asserted when I2C slave looses arbitration of the I2C bus, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[4]	I2C_SLV4_NACK	Asserted when slave 4 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[3]	I2C_SLV3_NACK	Asserted when slave 3 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[2]	I2C_SLV2_NACK	Asserted when slave 2 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[1]	I2C_SLV1_NACK	Asserted when slave 1 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.
[0]	I2C_SLV0_NACK	Asserted when slave 0 receives a nack, will cause an interrupt if bit I2C_MST_INT_EN in the INT_ENABLE register is asserted.

4.19 Register 55 – INT Pin / Bypass Enable Configuration

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	ACTL	1 – The logic level for INT pin is active low. 0 – The logic level for INT pin is active high.
[6]	OPEN	1 – INT pin is configured as open drain. 0 – INT pin is configured as push-pull.
[5]	LATCH_INT_EN	1 – INT pin level held until interrupt status is cleared. 0 – INT pin indicates interrupt pulse's is width 50µs.
[4]	INT_ANYRD_2CLEAR	1 – Interrupt status is cleared if any read operation is performed. 0 – Interrupt status is cleared only by reading INT_STATUS register
[3]	ACTL_FSYNC	1 – The logic level for the FSYNC pin as an interrupt is active low. 0 – The logic level for the FSYNC pin as an interrupt is active high.
[2]	FSYNC_INT_MODE_EN	1 – This enables the FSYNC pin to be used as an interrupt. A transition to the active level described by the ACTL_FSYNC bit will cause an interrupt. The status of the interrupt is read in the I2C Master Status register PASS_THROUGH bit. 0 – This disables the FSYNC pin from causing an interrupt.
[1]	BYPASS_EN	When asserted, the I2C_MASTER interface pins (ES_CL and ES_DA) will go into 'bypass mode' when the I2C master interface is disabled. The pins will float high due to the internal pull-up if not enabled and the I2C master interface is disabled.
[0]	-	Reserved

4.20 Register 56 – Interrupt Enable

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	-	Reserved
[6]	WOM_EN	1 – Enable interrupt for wake on motion to propagate to interrupt pin. 0 – function is disabled.

BIT	NAME	FUNCTION
[5]	-	Reserved
[4]	FIFO_OVERFLOW_EN	1 – Enable interrupt for FIFO overflow to propagate to interrupt pin. 0 – function is disabled.
[3]	FSYNC_INT_EN	1 – Enable FSYNC interrupt to propagate to interrupt pin. 0 – function is disabled.
[2]	-	Reserved
[1]	-	Reserved
[0]	RAW_RDY_EN	1 – Enable Raw Sensor Data Ready interrupt to propagate to interrupt pin. The timing of the interrupt can vary depending on the setting in register 36 I2C_MST_CTRL, bit [6] WAIT_FOR_ES. 0 – function is disabled.

4.21 Register 58 – Interrupt Status

Serial IF: R/C

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	-	Reserved
[6]	WOM_INT	1 – Wake on motion interrupt occurred.
[5]	-	Reserved
[4]	FIFO_OVERFLOW_INT	1 – FIFO Overflow interrupt occurred. Note that the oldest data is has been dropped from the FIFO.
[3]	FSYNC_INT	1 – FSYNC interrupt occurred.
[2]	-	Reserved
[1]	DMP_INT	This bit automatically sets to 1 when the DMP interrupt has been generated.
[0]	RAW_DATA_RDY_INT	1 – Sensor Register Raw Data sensors are updated and Ready to be read. The timing of the interrupt can vary depending on the setting in register 36 I2C_MST_CTRL, bit [6] WAIT_FOR_ES.

4.22 Registers 59 to 64 – Accelerometer Measurements

Name: ACCEL_XOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_XOUT_H	High byte of accelerometer X-axis data.

Name: ACCEL_XOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_XOUT_L	Low byte of accelerometer X-axis data.

Name: ACCEL_YOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_YOUT_H	High byte of accelerometer Y-axis data.

Name: ACCEL_YOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_YOUT_L	Low byte of accelerometer Y-axis data.

Name: ACCEL_ZOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_ZOUT_H	High byte of accelerometer Z-axis data.

Name: ACCEL_ZOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_ZOUT_L	Low byte of accelerometer Z-axis data.

4.23 Registers 65 and 66 – Temperature Measurement

Name: TEMP_OUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	TEMP_OUT_H	High byte of the temperature sensor output

Name: TEMP_OUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	TEMP_OUT_L	Low byte of the temperature sensor output To convert the output of the temperature sensor to degrees C please use the formula below: **TEMP_degC** = ((TEMP_OUT – RoomTemp_Offset)/Temp_Sensitivity) + 21degC

4.24 Registers 67 to 72 – Gyroscope Measurements

Name: GYRO_XOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	GYRO_XOUT_H	High byte of the X-axis gyroscope output

Name: GYRO_XOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	GYRO_XOUT_L	Low byte of the X-axis gyroscope output **GYRO_XOUT =** Gyro_Sensitivity * X_angular_rate Nominal Conditions: FS_SEL = 0 Gyro_Sensitivity = 131 LSB/(º/s)

Name: GYRO_YOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	GYRO_YOUT_H	High byte of the Y-axis gyroscope output

Name: GYRO_YOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	GYRO_YOUT_L	Low byte of the Y-axis gyroscope output **GYRO_YOUT =** Gyro_Sensitivity * Y_angular_rate Nominal Conditions: FS_SEL = 0 Gyro_Sensitivity = 131 LSB/(º/s)

Name: GYRO_ZOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	GYRO_ZOUT_H	High byte of the Z-axis gyroscope output

Name: GYRO_ZOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	GYRO_ZOUT_L	Low byte of the Z-axis gyroscope output **GYRO_ZOUT =** Gyro_Sensitivity * Z_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(º/s)

4.25 Registers 73 to 96 – External Sensor Data

EXT_SENS_DATA_00 – 23

Serial IF: READ

Reset value: 0x00

24 registers with the same description as below:

BIT	NAME	FUNCTION
[7:0]	EXT_SENS_DATA_xx	Sensor data read from external I2C devices via the I2C master interface. The data stored is controlled by the I2C_SLV(0-4)_ADDR, I2C_SLV(0-4)_REG, and I2C_SLV(0-4)_CTRL registers

Description:

These registers store data read from external sensors by the Slave 0, 1, 2, and 3 on the auxiliary I^2C interface. Data read by Slave 4 is stored in I2C_SLV4_DI (Register 53).

External sensor data is written to these registers at the Sample Rate as defined in Register 25. This access rate can be reduced by using the Slave Delay Enable registers (Register 103).

Data is placed in these external sensor data registers according to I2C_SLV0_CTRL, I2C_SLV1_CTRL, I2C_SLV2_CTRL, and I2C_SLV3_CTRL (Registers 39, 42, 45, and 48). When more than zero bytes are read (*I2C_SLVx_LEN* > 0) from an enabled slave (*I2C_SLVx_EN* = 1), the slave is read at the Sample Rate (as defined in Register 25) or delayed rate (if specified in Register 52 and 103). During each sample cycle, slave reads are performed in order of Slave number. If all slaves are enabled with more than zero bytes to be read, the order will be Slave 0, followed by Slave 1, Slave 2, and Slave 3.

Each enabled slave will have EXT_SENS_DATA registers associated with it by number of bytes read (*I2C_SLVx_LEN)* in order of slave number, starting from EXT_SENS_DATA_00. Note that this means enabling or disabling a slave may change the higher numbered slaves' associated registers. Furthermore, if fewer total bytes are being read from the external sensors as a result of such a change, then the data remaining in the registers which no longer have an associated slave device (i.e. high numbered registers) will remain in these previously allocated registers unless reset.

If the sum of the read lengths of all SLVx transactions exceed the number of available EXT_SENS_DATA registers, the excess bytes will be dropped. There are 24 EXT_SENS_DATA registers and hence the total read lengths between all the slaves cannot be greater than 24 or some bytes will be lost.

Note: Slave 4's behavior is distinct from that of Slaves 0-3. For further information regarding the characteristics of Slave 4, please refer to Registers 49 to 53.

Example:

Suppose that Slave 0 is enabled with 4 bytes to be read (*I2C_SLV0_EN* = 1 and *I2C_SLV0_LEN* = 4) while Slave 1 is enabled with 2 bytes to be read, (*I2C_SLV1_EN*=1 and *I2C_SLV1_LEN* = 2). In such a situation, EXT_SENS_DATA _00 through _03 will be associated with Slave 0, while EXT_SENS_DATA _04 and 05 will be associated with Slave 1.

If Slave 2 is enabled as well, registers starting from EXT_SENS_DATA_06 will be allocated to Slave 2.

If Slave 2 is disabled while Slave 3 is enabled in this same situation, then registers starting from EXT_SENS_DATA_06 will be allocated to Slave 3 instead.

Register Allocation for Dynamic Disable vs. Normal Disable

If a slave is disabled at any time, the space initially allocated to the slave in the EXT_SENS_DATA register, will remain associated with that slave. This is to avoid dynamic adjustment of the register allocation.

The allocation of the EXT_SENS_DATA registers is recomputed only when (1) all slaves are disabled, or (2) the *I2C_MST_RST* bit is set (Register 106).

This above is also true if one of the slaves gets NACKed and stops functioning.

4.26 Register 99 – I²C Slave 0 Data Out

I2C_SLV0_DO

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV0_DO	Data out when slave 0 is set to write

For further information regarding Slave 1 control, please refer to Registers 37 to 39.

4.27 Register 100 – I²C Slave 1 Data Out

I2C_SLV1_DO

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV1_DO	Data out when slave 1 is set to write

For further information regarding Slave 1 control, please refer to Registers 40 to 42.

4.28 Register 101 – I²C Slave 2 Data Out

I2C_SLV2_DO

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV2_DO	Data out when slave 2 is set to write

For further information regarding Slave 2 control, please refer to Registers 43 to 45.

4.29 Register 102 – I²C Slave 3 Data Out

I2C_SLV3_DO

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	I2C_SLV3_DO	Data out when slave 3 is set to write

For further information regarding Slave 2 control, please refer to Registers 46 to 48.

4.30 Register 103 – I²C Master Delay Control

I2C_MST_DELAY_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	DELAY_ES_SHADOW	Delays shadowing of external sensor data until all data is received
[6:5]	-	Reserved
[4]	I2C_SLV4_DLY_EN	When enabled, slave 4 will only be accessed (1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[3]	I2C_SLV3_DLY_EN	When enabled, slave 3 will only be accessed (1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[2]	I2C_SLV2_DLY_EN	When enabled, slave 2 will only be accessed 1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[1]	I2C_SLV1_DLY_EN	When enabled, slave 1 will only be accessed 1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG
[0]	I2C_SLV0_DLY_EN	When enabled, slave 0 will only be accessed 1+I2C_MST_DLY) samples as determined by SMPLRT_DIV and DLPF_CFG

4.31 Register 104 – Signal Path Reset

SIGNAL_PATH_RESET

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:3]	-	Reserved
[2]	GYRO_RST	Reset gyro digital signal path. Note: Sensor registers are not cleared. Use SIG_COND_RST to clear sensor registers.
[1]	ACCEL_RST	Reset accel digital signal path. Note: Sensor registers are not cleared. Use SIG_COND_RST to clear sensor registers.
[0]	TEMP_RST	Reset temp digital signal path. Note: Sensor registers are not cleared. Use SIG_COND_RST to clear sensor registers.

4.32 Register 105 – Accelerometer Interrupt Control

ACCEL_INTEL_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	ACCEL_INTEL_EN	This bit enables the Wake-on-Motion detection logic.
[6]	ACCEL_INTEL_MODE	This bit defines 1 = Compare the current sample with the previous sample. 0 = Not used.
[5:0]	-	Reserved

Please refer to the Wake-on-Motion Interrupt section of the MPU-6500 Product Specification for additional details.

4.33 Register 106 – User Control

Name: USER_CTRL

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7]	DMP_EN	1 – Enables DMP features. 0 – DMP features are disabled after the current processing round has completed.
[6]	FIFO_EN	1 – Enable FIFO operation mode. 0 – Disable FIFO access from serial interface. To disable FIFO writes by DMA, use FIFO_EN register. To disable possible FIFO writes from DMP, disable the DMP.
[5]	I2C_MST_EN	1 – Enable the I2C Master I/F module; pins ES_DA and ES_SCL are isolated from pins SDA/SDI and SCL/ SCLK. 0 – Disable I2C Master I/F module; pins ES_DA and ES_SCL are logically driven by pins SDA/SDI and SCL/ SCLK. NOTE: DMP will run when enabled, even if all internal sensors are disabled, except when the sample rate is set to 8kHz.
[4]	I2C_IF_DIS	1 – Reset I2C Slave module and put the serial interface in SPI mode only. This bit auto clears after one clock cycle of the internal 20MHz clock.
[3]	DMP_RST	This bit resets the DMP when set to 1 while DMP_EN equals 0. This bit automatically clears to 0 after the reset has been triggered.
[2]	FIFO_RST	1 – Reset FIFO module. Reset is asynchronous. This bit auto clears after one clock cycle of the internal 20MHz clock.
[1]	I2C_MST_RST	1 – Reset I2C Master module. Reset is asynchronous. This bit auto clears after one clock cycle of the internal 20MHz clock. NOTE: This bit should only be set when the I2C master has hung. If this bit is set during an active I2C master transaction, the I2C slave will hang, which will require the host to reset the slave.
[0]	SIG_COND_RST	1 – Reset all gyro digital signal path, accel digital signal path, and temp digital signal path. This bit also clears all the sensor registers.

4.34 Register 107 – Power Management 1

Name: PWR_MGMT_1

Serial IF: R/W

Reset value: 0x01

BIT	NAME	FUNCTION
[7]	DEVICE_RESET	1 – Reset the internal registers and restores the default settings. Write a 1 to set the reset, the bit will auto clear.
[6]	SLEEP	When set, the chip is set to sleep mode.
[5]	CYCLE	When set, and SLEEP and STANDBY are not set, the chip will cycle between sleep and taking a single sample at a rate determined by LP_ACCEL_ODR (MPU-6500 mode) or LP_WAKE_CTRL (MPU-6050 compatible mode) NOTE: When all accelerometer axis are disabled via PWR_MGMT_2 register bits and cycle is enabled, the chip will wake up at the rate determined by the respective registers above, but will not take any samples.
[4]	GYRO_STANDBY	When set, the gyro drive and pll circuitry are enabled, but the sense paths are disabled. This is a low power mode that allows quick enabling of the gyros.
[3]	TEMP_DIS	When set to 1, this bit disables the temperature sensor.
[2:0]	CLKSEL[2:0]	**Code** **Clock Source** 0 Internal 20MHz oscillator 1 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 2 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 3 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 4 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 5 Auto selects the best available clock source – PLL if ready, else use the Internal oscillator 6 Internal 20MHz oscillator 7 Stops the clock and keeps timing generator in reset

Note:

When using SPI interface, user should use PWR_MGMT_1 (register 107) as well as SIGNAL_PATH_RESET (register 104) to ensure the reset is performed properly. The sequence used should be:

1. Set H_RESET = 1 (register PWR_MGMT_1)
2. Wait 100ms
3. Set GYRO_RST = ACCEL_RST = TEMP_RST = 1 (register SIGNAL_PATH_RESET)
4. Wait 100ms

4.35 Register 108 – Power Management 2

Name: PWR_MGMT_2

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:6]	LP_WAKE_CTRL[1:0]	Specifies the frequency of wake-ups during Accelerometer Only Low Power Mode.
[5]	DISABLE_XA	1 – X accelerometer is disabled 0 – X accelerometer is on
[4]	DISABLE_YA	1 – Y accelerometer is disabled 0 – Y accelerometer is on
[3]	DISABLE_ZA	1 – Z accelerometer is disabled 0 – Z accelerometer is on
[2]	DISABLE_XG	1 – X gyro is disabled 0 – X gyro is on
[1]	DISABLE_YG	1 – Y gyro is disabled 0 – Y gyro is on
[0]	DISABLE_ZG	1 – Z gyro is disabled 0 – Z gyro is on

MPU-6050 compatible mode:

In addition to the above default MPU-6500 mode, this register allows the user to configure the frequency of wake-ups in Accelerometer Only Low Power Mode using the *LP_WAKE_CTRL* bits.

The MPU-6500 can be put into Accelerometer Only Low Power Mode using the following steps:

(i) Set CYCLE bit to 1
(ii) Set SLEEP bit to 0
(iii) Set TEMP_DIS bit to 1

(iv) Set DIS_XG, DIS_YG, DIS_ZG bits to 1

The bits mentioned in the steps (i) to (iii) can be found in Power Management 1 register (Register 107).

In this mode, the device will power off all devices except for the primary I^2C interface, waking only the accelerometer at fixed intervals to take a single measurement. The frequency of wake-ups can be configured with *LP_WAKE_CTRL* as shown below.

LP_WAKE_CTRL	Wake-up Frequency
0	1.25 Hz
1	5 Hz
2	20 Hz
3	40 Hz

4.36 Register 114 and 115 – FIFO Count Registers

Name: FIFO_COUNT_H

Address: 114

Serial IF: READ

Reset value: 0x00

BIT	NAME	FUNCTION
[7:5]	-	Reserved
[4:0]	FIFO_CNT[12:8]	High Bits, count indicates the number of written bytes in the FIFO. Reading this byte latches the data for both FIFO_COUNT_H, and FIFO_COUNT_L.

Name: FIFO_COUNT_L

Address: 115

Serial IF: READ

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	FIFO_CNT[7:0]	Low Bits, count indicates the number of written bytes in the FIFO. NOTE: Must read FIFO_COUNT_H to latch new data for both FIFO_COUNT_H and FIFO_COUNT_L.

4.37 Register 116 – FIFO Read Write

Name: FIFO_R_W

Serial IF: READ

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	D[7:0]	Read/Write command provides Read or Write operation for the FIFO.

Description:

This register is used to read and write data from the FIFO buffer.

Data is written to the FIFO in order of register number (from lowest to highest). If all the FIFO enable flags (see below) are enabled and all External Sensor Data registers (Registers 73 to 96) are associated with a Slave device, the contents of registers 59 through 96 will be written in order at the Sample Rate.

The contents of the sensor data registers (Registers 59 to 96) are written into the FIFO buffer when their corresponding FIFO enable flags are set to 1 in FIFO_EN (Register 35). An additional flag for the sensor data registers associated with I^2C Slave 3 can be found in I2C_MST_CTRL (Register 36).

If the FIFO buffer has overflowed, the status bit *FIFO_OFLOW_INT* is automatically set to 1. This bit is located in INT_STATUS (Register 58). When the FIFO buffer has overflowed, the oldest data will be lost and new data will be written to the FIFO unless register 26 CONFIG, bit[6] FIFO_MODE = 1.

If the FIFO buffer is empty, reading this register will return the last byte that was previously read from the FIFO until new data is available. The user should check *FIFO_COUNT* to ensure that the FIFO buffer is not read when empty.

4.38 Register 117 – Who Am I

Name: WHOAMI

Serial IF: READ

Reset value: 0x70

BIT	NAME	FUNCTION
[7:0]	WHOAMI	Register to indicate to user which device is being accessed.

This register is used to verify the identity of the device. The contents of *WHO_AM_I* is an 8-bit device ID. The default value of the register is 0x70 for MPU-6500. This is different from the I2C address of the device as seen on the slave I2C controller by the applications processor. The I2C address of the MPU-6500 is 0x68 or 0x69 depending upon the value driven on AD0 pin.

4.39 Registers 119, 120, 122, 123, 125, 126 Accelerometer Offset Registers

For MPU-6500 mode:

Name: XA_OFFS_H

Address: 119

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	XA_OFFS[14:7]	Upper bits of the X accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

Name: XA_OFFS_L

Address: 120

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:1]	XA_OFFS[6:0]	Lower bits of the X accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps
[0]	-	Reserved

Name: YA_OFFS_H

Address: 122

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	YA_OFFS[14:7]	Upper bits of the Y accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

Name: YA_OFFS_L

Address: 123

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION

BIT	NAME	FUNCTION
[7:1]	YA_OFFS[6:0]	Lower bits of the Y accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps
[0]	-	Reserved

Name: ZA_OFFS_H

Address: 125

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:0]	ZA_OFFS[14:7]	Upper bits of the Z accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps

Name: ZA_OFFS_L

Address: 126

Serial IF: R/W

Reset value: 0x00

BIT	NAME	FUNCTION
[7:1]	ZA_OFFS[6:0]	Lower bits of the Z accelerometer offset cancellation. +/- 16g Offset cancellation in all Full Scale modes, 15 bit 0.98-mg steps
[0]	-	Reserved

For MPU-6050 compatible mode:

In the MPU-6050 compatible mode, the registers mentioned above are located at registers 6 to 11.

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